



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering Academic Year 2022 – 2023 (Even Semester)

Degree, Semester & Branch: VIII Semester B.E.ECE A&B

Course Code & Title: GE8076 Professional Ethics in Engineering

Name of the Faculty member (s): Mrs.R.Ramalakshmi

Innovative Practice Description

- **Unit / Topic:** Unit II / Types of Enquiry, Moral Dilemmas, Moral Autonomy
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 6
- **Activity Chosen:** Flipped Class Room
- **Justification:**

The students will be able to understand the concepts of Types of Enquiry, Moral Dilemmas, Moral Autonomy with the help of Flipped Class Room activity more easily. The students will be able to easily remember the concepts by watching the videos before teaching the concept.

- **Time Allotted for the Activity:** 50 minutes

- **Details of the Implementation:**

A flipped classroom is an instructional strategy and a type of blended learning, which aims to increase student engagement and learning by having students complete readings at home and work on live problem-solving during class time.

The flipped classroom is a teaching approach where the students get their first exposure to course content before coming to class through readings and video lectures. They then spend in-class time engaging in activities that have been designed to promote a deeper understanding of a concept. The students learn the concepts of types of enquiry, moral dilemmas, moral autonomy by watching the videos before itself. Then the students were allowed to discuss the topics among their friends. Then students grouped and shared their views on the topic. Since the students discussed among their friends, they understood the concepts more clearly.

- **CO – PO / PSO mapping:**

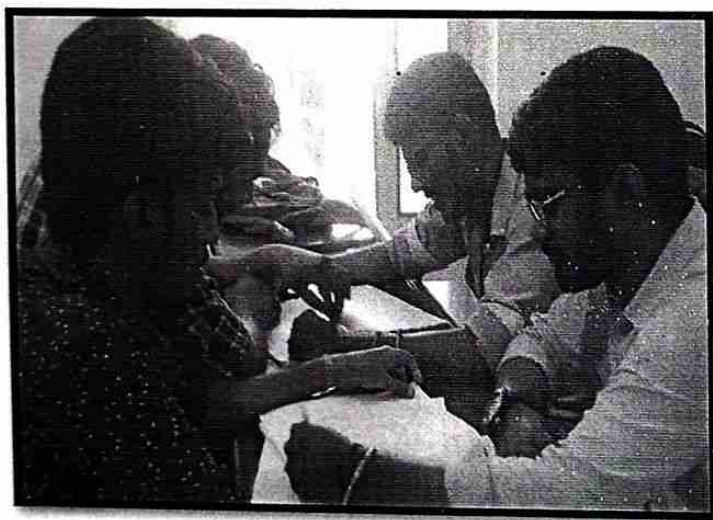
CO	PO6	PO8	PO9	PO12	PSO4
CO2	3	3	3	1	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO6	PO8	PO9	PO12	PSO4
	3	3	3	1	2
Justification for correlation	Flipped Class Room activity improves the learning performance of the students , there will be more interactions within the class room.	The students learn the concepts by watching videos or else through books previously by which there is an enhanced student satisfaction, engagement and enjoyment.	The activity helps the students to have a flexible learning environment as a team and also provides individualized learning	As the students do the activity, it will be more joyful by learning the concepts by themselves which will be a life long learning	This activity is really useful in learning the concepts of types of enquiry, moral dilemma, moral autonomy which will be utilized in contributing as entrepreneur.

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

The effectiveness of the activity was felt when the students were asked the questions on the topic, they were able to answer easily. They also understood the concepts more easily and able to remember while writing the test.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students were more interested in learning the concepts if they are sharing the

concepts through their friends. They clarified their doubts with the neighboring students and the faculty. They learnt the concepts and asked more doubts in the topic and learnt many things. Outcome attainment has also been increased since the students' performance in the test is also good.

❖ **Challenges faced in implementation:**

Immediate feedback cannot be got from the students through this activity. There will be a limited student preparation sometimes, more time and work for students.

Textbooks:

1. Mike W. Martin and Roland Schinzinger, "Ethics in Engineering", Tata Mc Graw Hill, New Delhi, 2003.
2. Govindarajan M, Natarajan S, Senthil Kumar V. S, "Engineering Ethics", Prentice Hall of India, New Delhi, 2004.

Other References:

1. Charles B. Fleddermann, -Engineering Ethicsl, Pearson Prentice Hall, New Jersey, 2004.
2. Charles E. Harris, Michael S. Pritchard and Michael J. Rabins, Engineering Ethics – Concepts and Cases, Cengage Learning, 2009.
3. John R Boatright, -Ethics and the Conduct of Businessl, Pearson Education, New Delhi, 2003
4. Edmund G Seebauer and Robert L Barry, -Fundamentals of Ethics for Scientists and Engineers, Oxford University Press, Oxford, 2001.
5. Laura P. Hartman and Joe Desjardins, -Business Ethics: Decision Making for Personal Integrity and Social Responsibility Mc Graw Hill education, India Pvt. Ltd., New Delhi, 2013.
6. World Community Service Centre, Value Education, Vethathiri publications, Erode, 2011.

Web sources:

1. www.onlineethics.org
2. www.nspe.org
3. www.globalethics.org



Signature of the Faculty Member



HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering Academic Year 2022 – 2023 (Even Semester)

Degree, Semester & Branch: VIII Semester B.E.ECE B
Course Code & Title: GE8076 Professional Ethics in Engineering
Name of the Faculty member (s): Mrs.R.Ramalakshmi

Innovative Practice Description

- **Unit / Topic:** Unit IV / Intellectual Property Rights
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 14
- **Activity Chosen:** Reciprocal Peer Questioning
- **Justification:**

The students will be able to understand the concepts of Intellectual Property Rights with the help of Reciprocal Peer Questioning activity more easily. The students will be able to easily remember the concepts if questions are being asked to them.

- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

Reciprocal Peer Questioning is an activity in which students question each other about the content they are learning using higher-order, open-ended question stems. The questions are used to promote thinking and generate focused discussions in small groups.

The students were in the class room. First the students were taught with the concepts of Intellectual Property Rights. Then the students were allowed to discuss the topics among their friends. Then the students were encouraged to ask questions to each other by which the concepts are learnt easily and more quickly.

- **CO – PO / PSO mapping:**

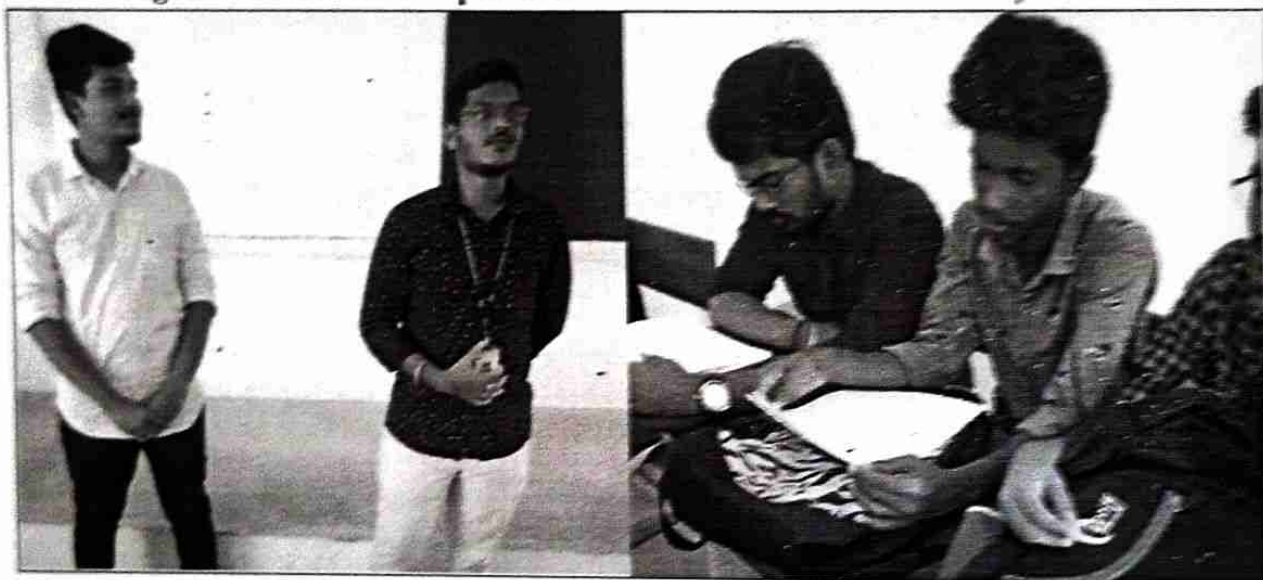
CO	PO1	PO2	PO3	PO6	PO7	PO8	PO9	PO12	PSO4
CO5	2	2	1	3	3	3	3	1	3

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO6	PO7	PO8	PO9	PSO4
				3	3
Justification for correlation	Reciprocal Questioning activity improves the performance of the students, there will be more interactions within the class room.	Peer This activity helps the students to actively participate for the environment and related activities.	The students get the knowledge on Intellectual Property Rights and involve in the activities ethically, morally and follow the principles in the society.	Reciprocal Questioning activity on IPR helps the students to present results as a team, with smooth integration of contributions from all individual efforts.	Peer The knowledge of Intellectual Property Rights is useful in contributing as entrepreneur.

• Images / Screenshot of the practice:



• Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

The effectiveness of the activity was felt when the students were asked the questions on the topic, they were able to answer easily. They also understood the concepts more easily and able to remember while writing the test.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students were more interested in learning the concepts if they are asking questions and sharing the concepts through their friends. They clarified their doubts with the neighboring students and the faculty. They learnt the concepts and asked more doubts in the topic and learnt many things. Outcome attainment has also been increased since the students' performance in the test is also good.

❖ **Challenges faced in implementation:**

Basically some time is needed to explain the basic concepts of Intellectual Property Rights to the students and also to discuss among the students.

Textbooks:

1. Mike W. Martin and Roland Schinzinger, "Ethics in Engineering", Tata Mc Graw Hill, New Delhi, 2003.
2. Govindarajan M, Natarajan S, Senthil Kumar V. S, "Engineering Ethics", Prentice Hall of India, New Delhi, 2004.

Other References:

1. Charles B. Fleddermann, -Engineering EthicsI, Pearson Prentice Hall, New Jersey, 2004.
2. Charles E. Harris, Michael S. Pritchard and Michael J. Rabins, Engineering Ethics – Concepts and Cases, Cengage Learning, 2009.
3. John R Boatright, -Ethics and the Conduct of BusinessI, Pearson Education, New Delhi, 2003
4. Edmund G Seebauer and Robert L Barry, -Fundamentals of Ethics for Scientists and Engineers, Oxford University Press, Oxford, 2001.
5. Laura P. Hartman and Joe Desjardins, -Business Ethics: Decision Making for Personal Integrity and Social Responsibility Mc Graw Hill education, India Pvt. Ltd., New Delhi, 2013.
6. World Community Service Centre, Value Education, Vethathiri publications, Erode, 2011.

Web sources:

1. www.onlineethics.org
2. www.nspe.org
3. www.globalethics.org


Signature of the Faculty Member

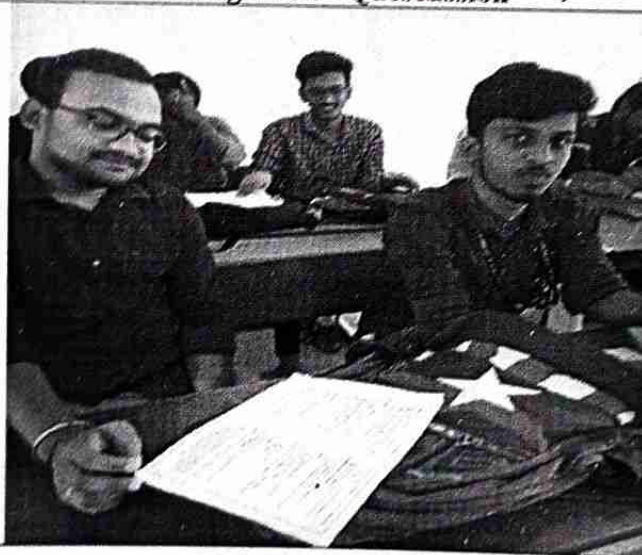


HOD

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2022 - 2023 (Even Semester)

INNOVATIVE TEACHING METHOD

Degree, Semester & Branch : VIII Semester B.E. ECE A&B
Course Code & Title : GE8076 Professional Ethics in Engineering
Name of the Faculty member: Mrs.R.Ramalakshmi

<u>Unit I</u>	
Cooperation, Commitment, Empathy - One minute speech	
	
Introduction to Meditation for professional excellence and stress management - Quescussion	
	

Unit II	
Professions and Professionalism, Models of professional roles - One Minute Paper	
	

Unit III	
Learning from the past - Three step interview	
	
Challenger Case Study - Brainstorming	
	

Unit IV

Risk Benefit Analysis – Think Pair Share

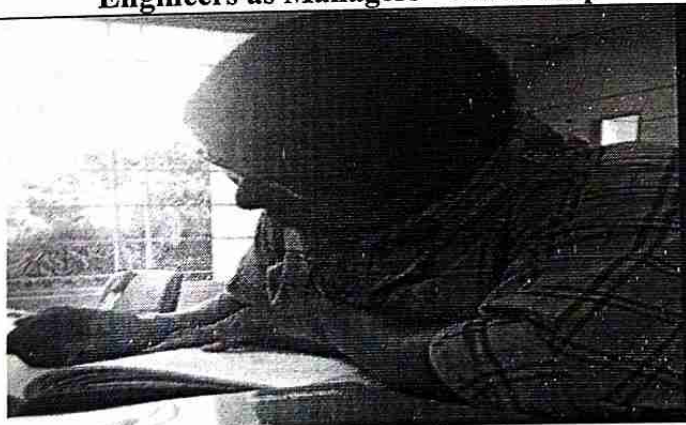


Unit V

Computer Ethics – Quiz Dominos



Engineers as Managers – Mind Map




Signature of the Faculty Member


HOD